ModelIt!

Teacher's Guide

Designing the Perfect Artificial Organ

Modeling the Bioengineering Challenge of Replacing Living Tissue with Technology

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-LS1-6, HS-PS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-6, HS-PS1-3

Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules. Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of immune rejection mechanisms and design bioengineering solutions that address biocompatibility, structural, and functional requirements simultaneously.
Disciplinary Core Idea	**LS1.A Structure and Function / PS1.A Structure and Properties of Matter** — Students analyze how the molecular structure and surface properties of implant materials determine biological function — how atomic-level surface chemistry controls protein adsorption, cell behavior, and immune recognition.
Crosscutting Concept	**Structure and Function** — Students model how the structure of an artificial organ at multiple scales — molecular surface, material bulk, device architecture — determines its biological function and survival in the body.

CAST Testing Alignment

MULTIPLE CHOICE:

An artificial heart valve made of pyrolytic carbon has excellent mechanical durability but triggers moderate immune activation. A new polymer valve has lower immune response but degrades after 8 years. According to your model, which variable ultimately limits Functional Lifespan for each design?

CONSTRUCTED RESPONSE:

Using your model, explain why surface chemistry modifications are more effective at reducing immune rejection than choosing a stronger bulk material. Address the molecular mechanisms of protein adsorption and immune cell recognition in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The human body contains approximately 78 organs, each performing functions so complex that replicating them artificially remains at the frontier of engineering. The kidney alone contains 1 million nephrons, each filtering blood through a membrane so precise it allows water and waste through while retaining blood cells and proteins. Replicating this with synthetic materials while avoiding immune rejection is a challenge that spans chemistry, biology, materials science, and mechanical engineering.

The Foreign Body Response

When any material enters the body, proteins immediately coat its surface within seconds (the Vroman effect). The immune system reads these adsorbed proteins to determine friend or foe. If the surface is recognized as foreign, macrophages arrive to engulf it. When they cannot (the device is too large), they fuse into giant cells and signal chronic inflammation. Fibroblasts then deposit collagen around the device, creating a fibrous capsule that isolates it from surrounding tissue. This capsule blocks blood flow integration, nutrient transport, and cell attachment — slowly strangling the device's function.

Surface Engineering Strategies

Modern bioengineering fights immune rejection primarily through surface chemistry. Polyethylene glycol (PEG) coatings create a hydration layer that prevents protein adsorption, making the surface nearly invisible to the immune system. Bioactive peptide coatings (like RGD sequences) actively encourage beneficial cell attachment while discouraging immune cell activation. Drug-eluting surfaces release anti-inflammatory molecules locally without systemic immunosuppression. Zwitterionic coatings mimic cell membrane chemistry to appear 'self' to immune surveillance.

The Vascularization Challenge

Living cells cannot survive more than 100-200 micrometers from a blood vessel — beyond this distance, oxygen and nutrients cannot diffuse fast enough to sustain metabolism. An artificial organ incorporating living cells must develop a vascular network within days of implantation or those cells die. Strategies include pre-vascularization (growing blood vessel networks in the device before implantation), angiogenic factor release (signaling the body to grow new vessels into the device), and microfluidic channels (engineering artificial capillary networks into the device structure).

Current Artificial Organ Successes and Limitations

Mechanical heart valves last 20-30 years but require lifelong blood-thinning medication. Hip and knee replacements last 15-25 years with excellent function. Cochlear implants restore hearing with remarkable neural integration. But fully artificial kidneys, livers, lungs, and pancreases remain elusive because these organs perform complex biochemical functions that synthetic materials cannot replicate — they require living cells, which in turn require vascularization, nutrients, and immune tolerance. The emerging field of bioartificial organs combines synthetic structures with living cells to bridge this gap.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the artificial organ system: biocompatible material, molecular surface chemistry, immune response rate, blood flow integration, structural durability, nutrient transport, waste filtering, electrical signal compatibility, cell attachment rate, and functional lifespan.

E **Establish Relationships**

Students map relationships showing that biocompatible material and surface chemistry jointly determine immune response rate, which controls blood flow integration and cell attachment, which together with structural durability determine functional lifespan.

V **Visualize & Model**

Students build a computational model showing the ten-component bioengineering system with immune rejection as the central failure pathway.

E **Evaluate Outcomes**

Students run optimal biocompatibility, immune rejection cascade, and mechanical failure scenarios to identify which variables most commonly limit artificial organ lifespan.

R **Revise & Extend**

Students add drug elution rate, degradation products, or infection resistance to model more realistic implant dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, 100,000 people in America are waiting for an organ transplant. 17 will die today because a donor organ will not arrive in time. What if we could build the organs ourselves?"*

Watch For

Share the transplant waiting list statistics. Show the gap between organs needed and organs available. Let the human stakes sink in before revealing the engineering challenge.

Slide 2: Learning Objectives

3 min

Say: *"Today you are engineering a device that must survive inside the most hostile environment for technology: the human body — warm, wet, corrosive, and defended by an immune system that wants to destroy anything foreign."*

Watch For

Have students read objectives. Pre-teach 'biocompatibility' and 'immune rejection.' Use analogy: the immune system is a security guard that attacks anything without the right ID badge.

Slide 3: Big Question

3 min

Say: *"We can build skyscrapers, rockets, and quantum computers. But we cannot build a replacement kidney that lasts a lifetime without the body attacking it. Why is biology harder than physics?"*

Watch For

Quick-write: Students list reasons they think the body might reject an artificial organ. Share out. Most will focus on material mismatch — set up the surface chemistry surprise.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model with ten variables spanning materials science, immunology, vascular biology, and mechanical engineering. This is what real biomedical engineering looks like."*

Watch For

Preview LEVER steps. Emphasize the interdisciplinary nature — this model requires understanding of biology AND engineering simultaneously.

Slide 5: Activity 1: Components

8 min

Say: *"Ten variables. Two inputs you control through material and surface design. Eight biological and mechanical responses. The body decides whether your design lives or dies."*

Watch For

Guide students through all ten components. Spend extra time on immune response rate — this is the gatekeeper variable that controls most other outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"The immune system attacks. Blood vessels do not grow in. Cells do not attach. Nutrients cannot reach the device. It dies. That is the rejection cascade — and surface chemistry is the only thing standing between your device and destruction."*

Watch For

Help students map the immune rejection cascade. Show how surface chemistry creates a protective barrier. Use green for integration pathways and red for rejection pathways.

Slide 7: Activity 3: Simulation

10 min

Say: *"Can your artificial organ survive 20 years inside the body? Run the scenarios and find out what kills it."*

Watch For

Students predict Functional Lifespan for each scenario. Run simulations. The immune rejection cascade should show how quickly a device fails once the immune response crosses a threshold.

Slide 8: Discoveries

5 min

Say: *"The body is the ultimate quality control system. It will test every square nanometer of your implant surface and destroy anything that fails the test. Your job is to pass the test."*

Watch For

Connect model findings to real artificial organs: mechanical heart valves, hip replacements, cochlear implants. What makes the successful ones work? Where do the failures happen?

Slide 9: STEM Challenge

12 min

Say: *"800,000 Americans need a kidney. 500,000 are on dialysis. Design the device that could save them — and make dialysis obsolete."*

Watch For

Groups design bioartificial kidney concepts. Present with engineering rationale for material choice, surface chemistry, vascularization strategy, and projected lifespan. Class evaluates feasibility.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Biocompatible Material, Molecular Surface Chemistry
- **INTERNAL:** Immune Response Rate, Blood Flow Integration, Structural Durability, Nutrient Transport, Waste Filtering, Electrical Signal Compatibility, Cell Attachment Rate, Functional Lifespan

Relationships

- **Molecular Surface Chemistry to Immune Response Rate:** NEGATIVE (-) — Better surface chemistry modifications (PEG coatings, zwitterionic surfaces, bioactive peptides) reduce the immune system's recognition and attack of the implant. Optimized surfaces can reduce immune response by 70-90% compared to unmodified materials.
- **Immune Response Rate to Blood Flow Integration:** NEGATIVE (-) — Higher immune response causes inflammation and fibrous capsule formation around the implant, blocking blood vessel growth into or around the device. Without vascularization, the implant becomes isolated from the body's circulatory system, starving any living cells within it.
- **Blood Flow Integration to Functional Lifespan:** POSITIVE (+) — Better blood flow integration provides nutrients and oxygen to living cells within the device, removes metabolic waste, and maintains the chemical environment needed for long-term function. Devices with poor vascularization suffer progressive cell death and functional decline.

Key Discoveries

- The immune system is both the greatest ally (protecting against infection at the implant site) and greatest enemy (attacking the implant itself) of artificial organ engineering
- Surface chemistry is more important than bulk material properties for immune compatibility — the body interacts with the nanometer-thick surface layer, not the interior material
- Functional lifespan is limited by whichever variable fails first — a mechanically perfect implant that triggers immune rejection fails just as surely as an immunologically compatible implant that fractures
- The ideal artificial organ would be invisible to the immune system, integrated with the circulatory and nervous systems, and durable enough to last a lifetime — no current technology achieves all three simultaneously

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The body rejects implants because they are the wrong shape or size"	Immune rejection is primarily a molecular surface phenomenon, not a geometric one. A perfectly shaped implant made of an immunogenic material will be rejected, while an oddly shaped implant with excellent surface chemistry will be accepted. The immune system does not care about the device's form — it cares about the molecular signals on its surface. Proteins that adsorb onto the surface within seconds of implantation determine the immune response. Shape matters for mechanical function, but surface chemistry determines biological survival.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Artificial organs just need to be made of something the body cannot dissolve"	Durability against chemical dissolution is necessary but not sufficient. Many biologically inert materials (stainless steel, some ceramics) resist degradation but trigger severe immune responses because their surfaces activate protein adsorption and immune cell recognition. Conversely, some biodegradable materials (certain hydrogels, PLGA polymers) deliberately dissolve over time but are designed to be replaced by the patient's own tissue. The goal is not permanence but functional compatibility — some of the best implant strategies involve temporary scaffolds that guide tissue regeneration and then disappear.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We will just 3D-print replacement organs soon"	3D bioprinting is promising but faces enormous challenges. Printing the structure of an organ is achievable — printing a structure with living cells that organize into functional tissue, develop vascular networks, and connect to the body's circulatory and nervous systems is vastly harder. The liver alone contains 500+ different cell types arranged in precise spatial patterns. Current bioprinting can create simple tissues (skin, cartilage) but complex organs with internal vasculature, innervation, and multiple cell types remain years to decades away. The model shows why: Cell Attachment Rate, Blood Flow Integration, and Nutrient Transport must all work simultaneously at the microscale.